OPTIMIZER PERFORMANCE ON NON-CONVEX FUNCTIONS

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1. Objective

- We are checking the convergence behaviour and time taken by each optimizer- **Gradient Decent, SGD with momentum, Adam, RMSprop, Adagrad** for optimizing **Rosenbrock function** and **Ackley function (1D simplified)**.

- Also the impact of hyperparametres on convergence speed and the final result.

2. The Non-convex Functions

- **Rosenbrock function:**

$$f(x, y) = (a - x)^2 + b(y - x^2)^2,$$

  where $a = 1$ and $b = 100$.

- **Ackley function (1D simplified):**

$$f(x) = -20\exp(-0.2\sqrt{x^2}) - \exp(\cos(2\pi x)) + 20 + e.$$

  Handle $x = 0$ by setting $f(0)$ or slightly exceeding zero where applicable to avoid domain errors.

3. Experimental Setup

We are using the following optimizer :

- Gradient Descent

- Stochastic Gradient Descent with momentum

- Adam
- RMSprop
- Adagrad.

with following learning rates(α) :

$$0.01, \qquad 0.05, \qquad 0.1$$

4. Results For Rosenbrock

Threshold criteria - $1e^{-}8$

- <u>Gradient Descent</u>
  - For $\alpha = 0.01$, algorithm terminates at 7th iteration with **nan**. Time = 0.011284112930297852 .
  - For $\alpha = 0.05$, algorithm terminates at 6th iteration with **nan**. Time =0.0023725032806396484 .
  - For $\alpha = 0.1$, algorithm terminates at 6th iteration with **nan**. Time =0.00026726722717285156 .
- <u>SGD with Momentum</u>
  - For $\alpha = 0.01$, algorithm terminates at 7th iteration with **nan**. Time =0.0027196407318115234
  - For $\alpha = 0.05$, algorithm terminates at 6th iteration with **nan**. Time = 0.000331878662109375
  - For $\alpha = 0.1$, algorithm terminates at 6th iteration with **nan**. Time =0.0002474784851074219

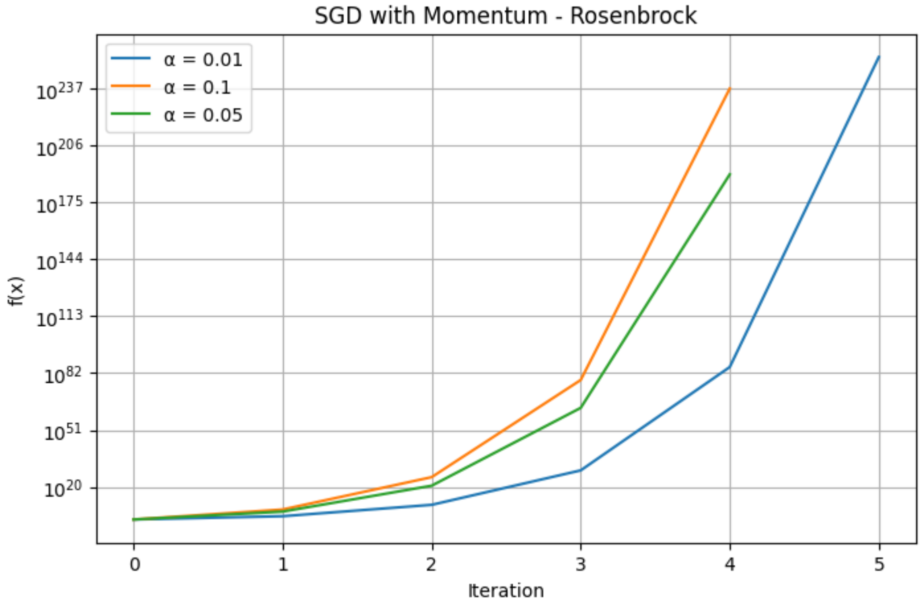

Figure 1: SGD with Momentum - Rosenbrock

- <u>Adam</u>
  - For $\alpha = 0.01$, algorithm terminates at 8765th iteration with 1.2108894536940508e-16. Time=0.3037395477294922
  - For $\alpha = 0.05$, algorithm terminates at 4529th iteration with 1.1750519154239464e-16. Time=0.141527414321899941
  - For $\alpha = 0.1$, algorithm terminates at 3202th iteration with 1.1439635353673058e-16. Time=0.204363584518432262

- <u>RMSprop</u>
  - For $\alpha = 0.01$, algorithm run full iteration with functional value 0.0224507448047054443 . Time=0.2583789825439453
  - For $\alpha = 0.05$, algorithm run full iteration with functional value 0.46562499947565883. Time=0.32065868377685547

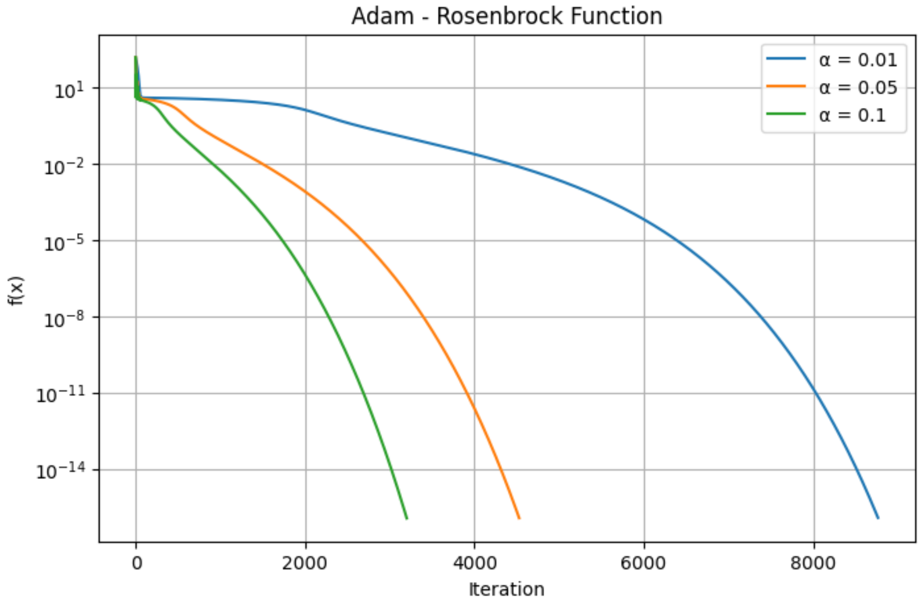

Figure 2: Adam - Rosenbrock

- For $\alpha = 0.1$, algorithm run full iteration with functional value 1.20249999881886. Time= 0.30153942108154297

- Adagrad

  - For $\alpha = 0.1$, algorithm run full iteration(10k) with functional value 0.10427161944264576. Time=0.2374422550201416
  - For $\alpha = 0.05$, algorithm run full iteration(10k) with functional value 1.988938330069468. Time= 0.28369569778442383
  - For $\alpha = 0.01$, algorithm run full iteration(10k) with functional value 4.009592760226259. Time=0.37450218200683594

- **We are not compare GD and SGD-M as they give infinte value at starting iteraion at given learning rates.**

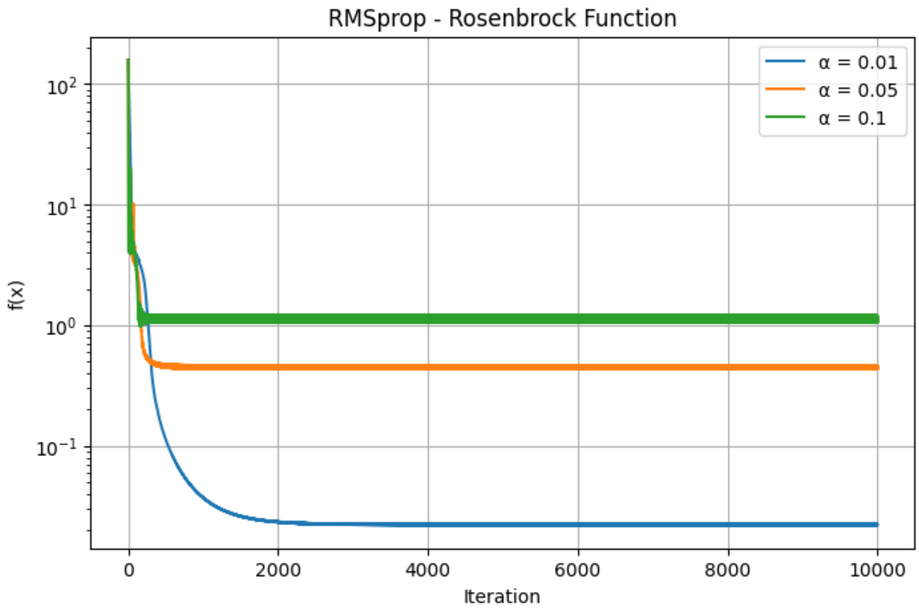

Figure 3: RMSprop - Rosenbrock

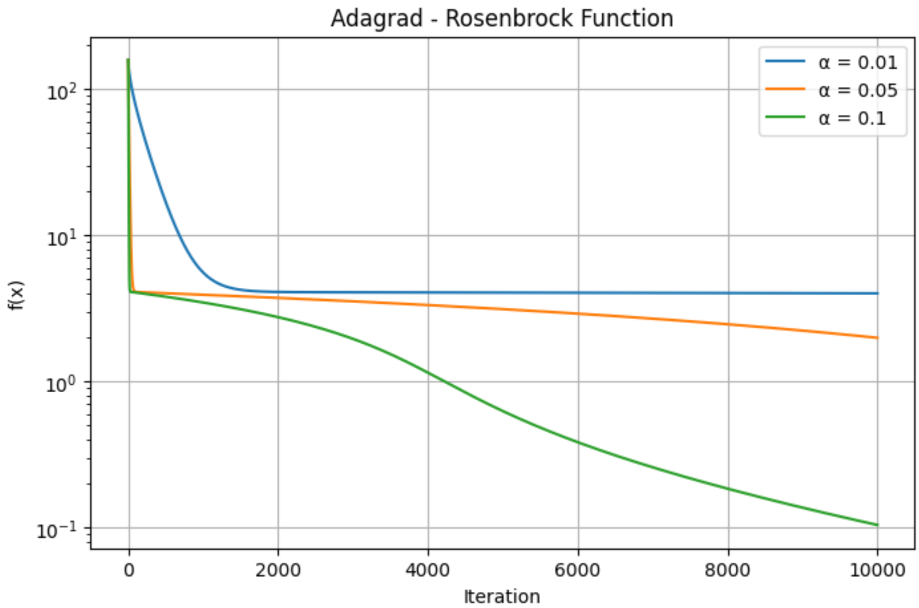

Figure 4: Adagrad - Rosenbrock

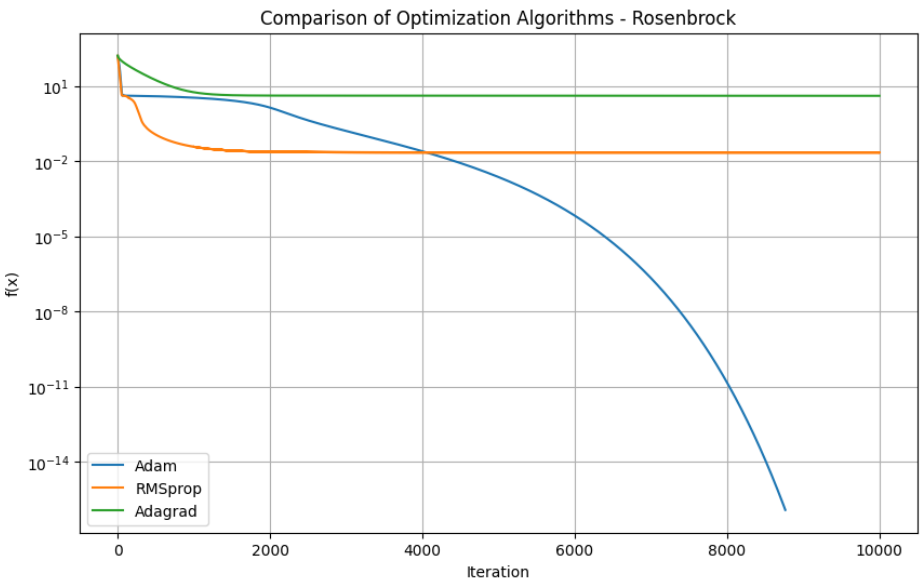

Figure 5: Comparison of 3 Algorithm: $\alpha = 0.01$

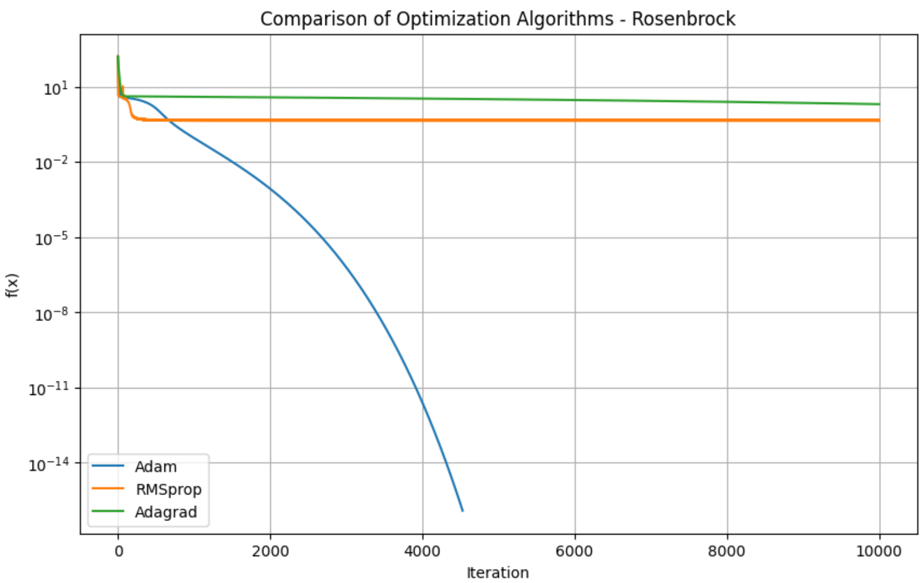

Figure 6: Comparison of 3 Algorithm: $\alpha = 0.05$

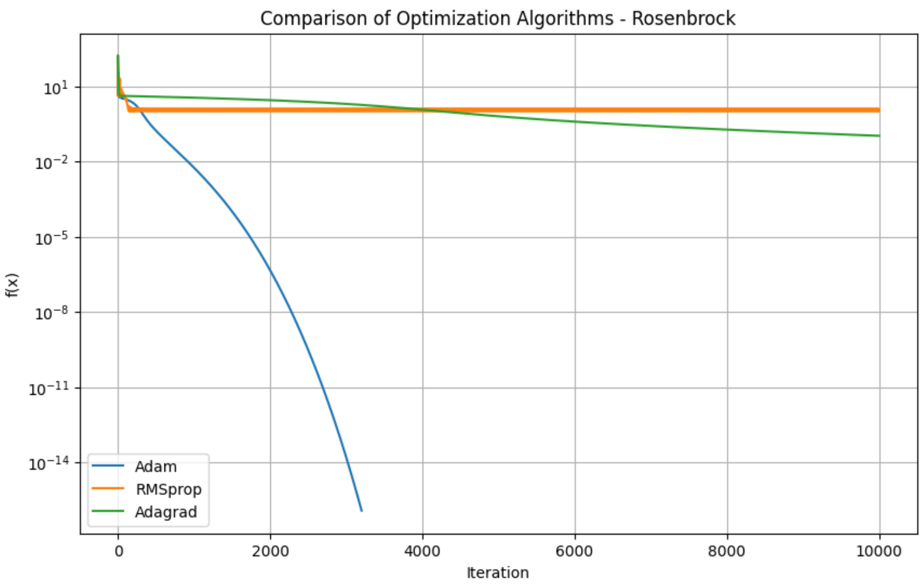

Figure 7: Comparison of 3 Algorithm: $\alpha = 0.1$

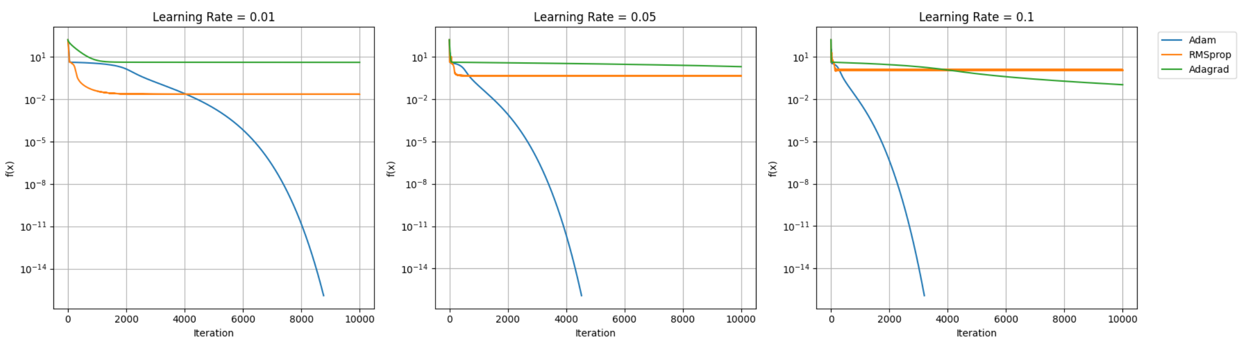

Figure 8: Comparison of 3 LR with algorithm

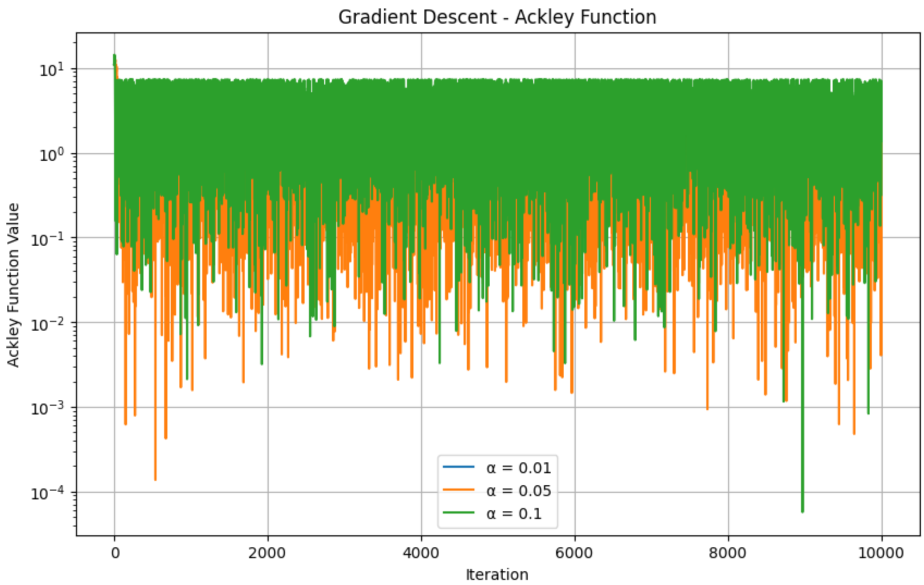

Figure 9: Gradient Descent - Ackley

5. Results For Ackley

Threshold criteria - $1e^{-}8$

- Gradient Descent
  - For $\alpha = 0.01$, algorithm terminates at 8th iteration with functional value 10.998262102599448. Time = 0.00020962000053259544 .
  - For $\alpha = 0.05$, algorithm run full iteration(10k) with functional value 3.7426601953693894. Time =0.11620480599958682 .
  - For $\alpha = 0.1$, algorithm run full iteration(10k) with functional value 7.093118704709461. Time =0.2324354750016937 .
- SGD with Momentum

Figure 10: SGDM - Ackley

- For $\alpha = 0.01$, algorithm terminates at 318th iteration with functional value 10.998262102599446. Time = 0.004882171999270213.
- For $\alpha = 0.05$, algorithm run full iteration(10k) with functional value 20.61108352458559. Time =0.14439763500013214.
- For $\alpha = 0.1$, algorithm run full iteration(10k) with functional value 20.26188770239447. Time =0.14694794999923033.

• <u>Adam</u>

- For $\alpha = 0.01$, algorithm terminates at 294th iteration with functional value 10.998262102599444. Time = 0.0055326969995803665.
- For $\alpha = 0.05$, algorithm terminates at 338th iteration with functional value 10.998262102599446. Time =0.005516840999916894.
- For $\alpha = 0.1$, algorithm terminates at 327th iteration with func-

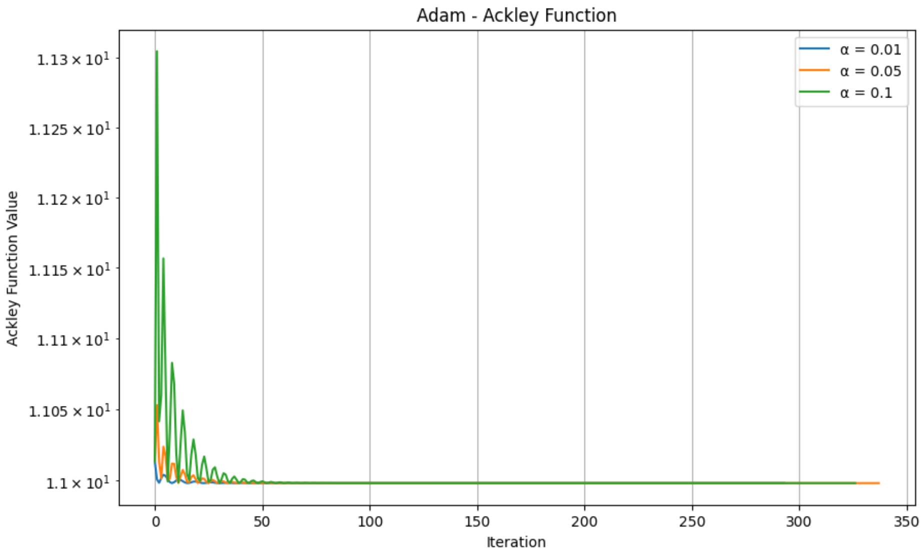

Figure 11: Adam - Ackley

tional value 10.998262102599444. Time =0.004752046001158305 4.

- RMSprop
  - For $\alpha = 0.01$, algorithm run full iteration(10k) with functional value 10.999556586361898. Time = 0.1441723460011417.
  - For $\alpha = 0.05$, algorithm run full iteration(10k) with functional value 11.0291130733158. Time =0.150962605999666 27.
  - For $\alpha = 0.1$, algorithm run full iteration(10k) with functional value 11.137522799437392. Time =0.15334508999876562.

- Adagrad
  - For $\alpha = 0.01$, algorithm terminates at 26th iteration with functional value 10.998262102599446. Time=0.0012355510007182602.

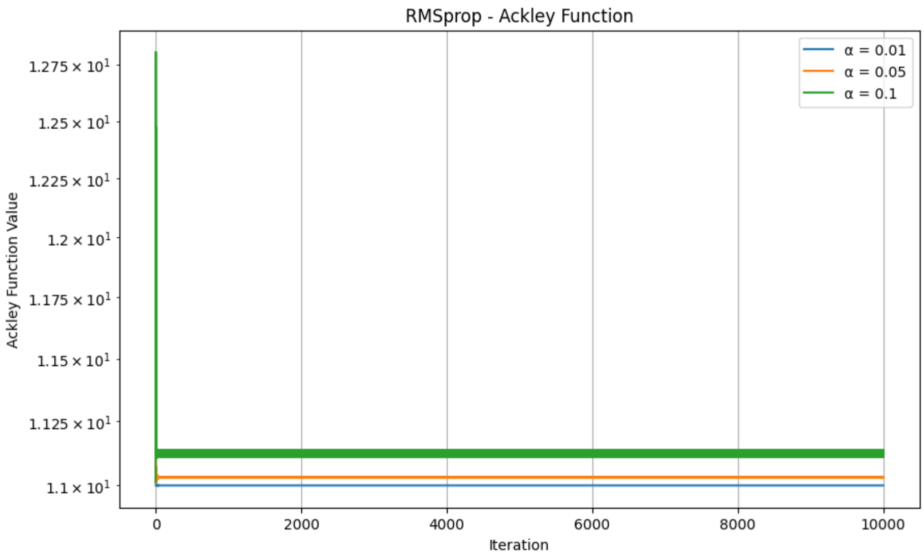

Figure 12: RMSprop - Ackley

- For $\alpha = 0.05$, algorithm terminates at 21th iteration with functional value 10.998262102599446. Time =0.0015722030002507381.
- For $\alpha = 0.1$, algorithm terminates at 32th iteration with functional value 10.998262102599446. Time =0.0005334799989213934.

Conclusion

- For Rosenbrock function, out of all five Adam algorithm work well and GD, SGD-Momentum are worst.
- For Ackley function, out of all five algorithm GD with $\alpha = 0.05$ work well and others are premature terminates at these give learning rates of $0.01, 0.05, 0.1$.

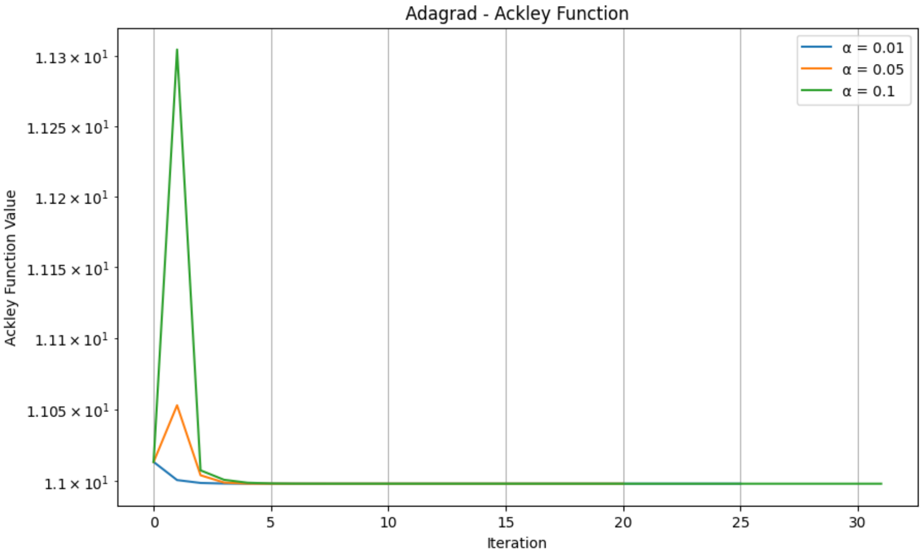

Figure 13: Adagrad - Ackley

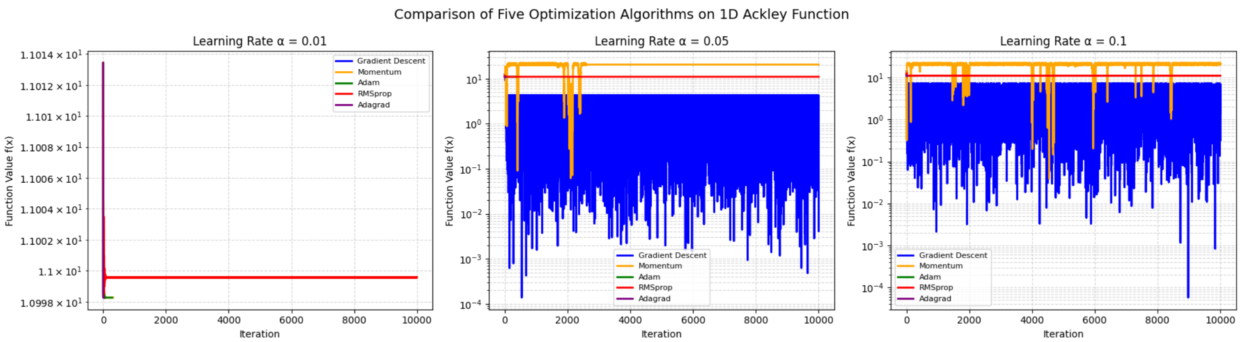

Figure 14: Comparision of five algorithms

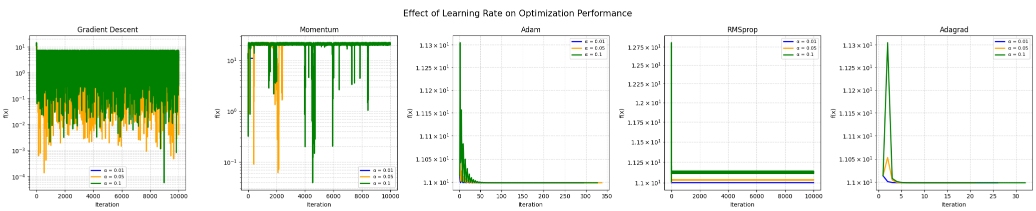

Figure 15: Effect of LR on optimizer